GALGOTIAS COLLEGE OF ENGINEERING & TECHNOLOGY

Object Oriented Programming Lab

Subject Code: BMC252

Semester: II

Session: 2025-26

Submitted To:

Mr Md Shahajad
AP, MCA Department

Submitted By:

Anshika Trivedi
Roll No: 2500970140031

Department of Computer Applications

Knowledge Park I, Greater Noida, Uttar Pradesh 201310

INDEX

S.N	Name of the Program	Date of Implementation	Date of Submission	Sign
1	Write a program in Java to print Hello.			
2	Write a program in java to understand the difference between print () and println ()			
3	Write a program in Java with two classes create a object of first class and call into another class (having main method)			
4	Write a program in java to product of two numbers			
5	Write a program in java product of two numbers (Input by the user)			
6	Write a program in java to illustrate the concept of local, instance and static variable			
7	Write a program in java to implement implicit and explicit type casting			
8	Write a program in java to implement the various operators in java			
9	Write a program in Java for constructor overloading			
10	Write a program in Java for method overloading			
11	Write a program in Java for method overriding			
12	Write a program in java to show run time polymorphism (up casting)			
13	Write a program in java for access specifiers (all four)			
14	Write a program in java to implement the single dimension array			
15	Write a program in java to copy the elements from one array to another array			
16	Write a program in java to perform the addition and multiplication in 2-D array			
17	Write a program in Java for simple inheritance			
18	Write a program in java for Final keyword			
19	Write a program in java for super keyword			
20	Write a program in Java chaining constructor			
21	Write a program in Java for abstract method and abstract class			
22	Write a program in Java for interface			
23	Write a program in Java multiple inheritance			
24	Write a program in Java for Object Cloning (shallow and deep copy)			
25	Write a program in Java for Inner Classes (all types)			
26	Write a program in java to create the package (user defined package)			
27	Write a program in Java for exception handling by using try, catch and finally			

28	Write a program in Java for throw and throws Exception			
29	Write a program in Java to throw your own Exceptions			
30	Write a program in Java to reading and writing in file using byte stream			
31	Write a program in Java to reading and writing in file using character stream.			
32	Write a program in Java to reading and writing through console class.			
33	Write a program in Java how to create thread using Thread Class.			
34	Write a program in Java how to create thread using runnable interface.			
35	Write a program in Java to implement the multithreading.			
36	Write a program in Java to achieve synchronization in threads.			
37	Write a program in Java to implement the concept of Priorities in threads.			
38	Write a program in Java to illustrate the concept of Generic Programming			
39	Write a program in Java to illustrate the concept of event handling (using various event handlers)			
40	Create a simple student registration application using various swing components (like: JFrame, JButton, JLabel, text fields, text areas, check box and radio buttons).			

Program No. 1

Objective: Write a program in Java to print Hello.

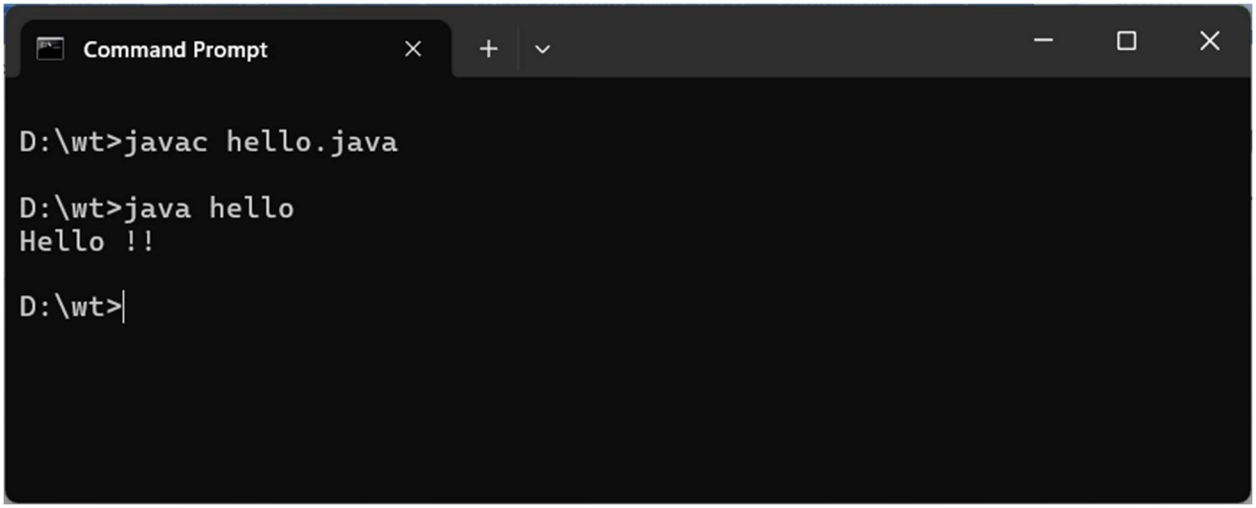
Code Implementation (Source Code):

```
public class Main {
    public static void main(String[] args) {
        System.out.println("Hello !!");
    }
}
```

Output:

```
Hello, World!
```

Output Screen:



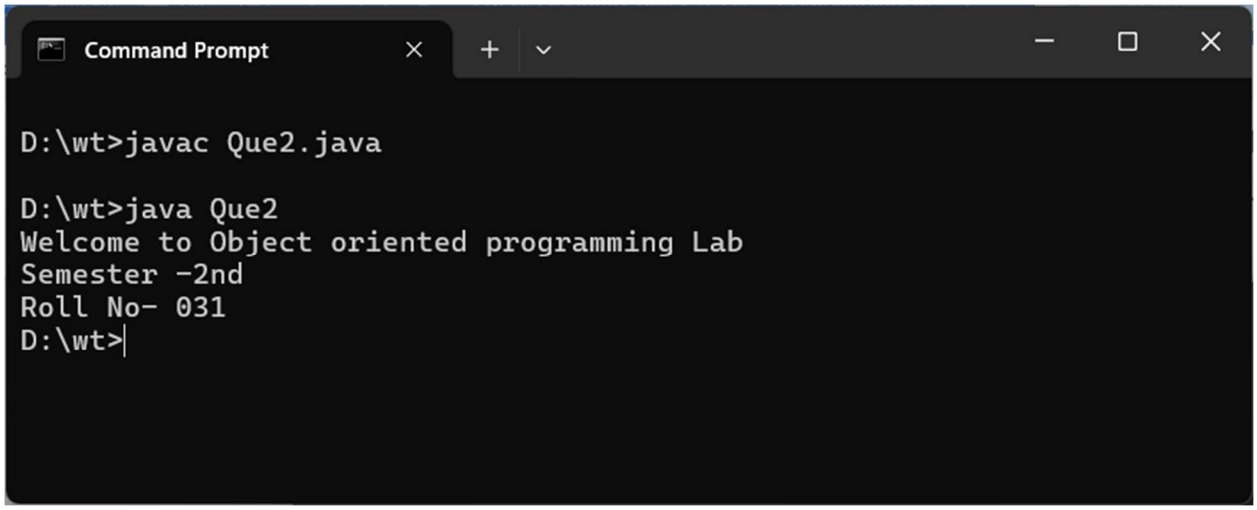
Program No. 2

Objective: Write a program in java to understand the difference between print () and println ().

Code Implementation (Source Code):

```
public class Que2{

public static void main(String [] args){

System.out.println("Welcome to Object oriented programming Lab");
System.out.println("Semester -2nd");
System.out.print(" Roll No-031“);

}
}
```

Output Screen:



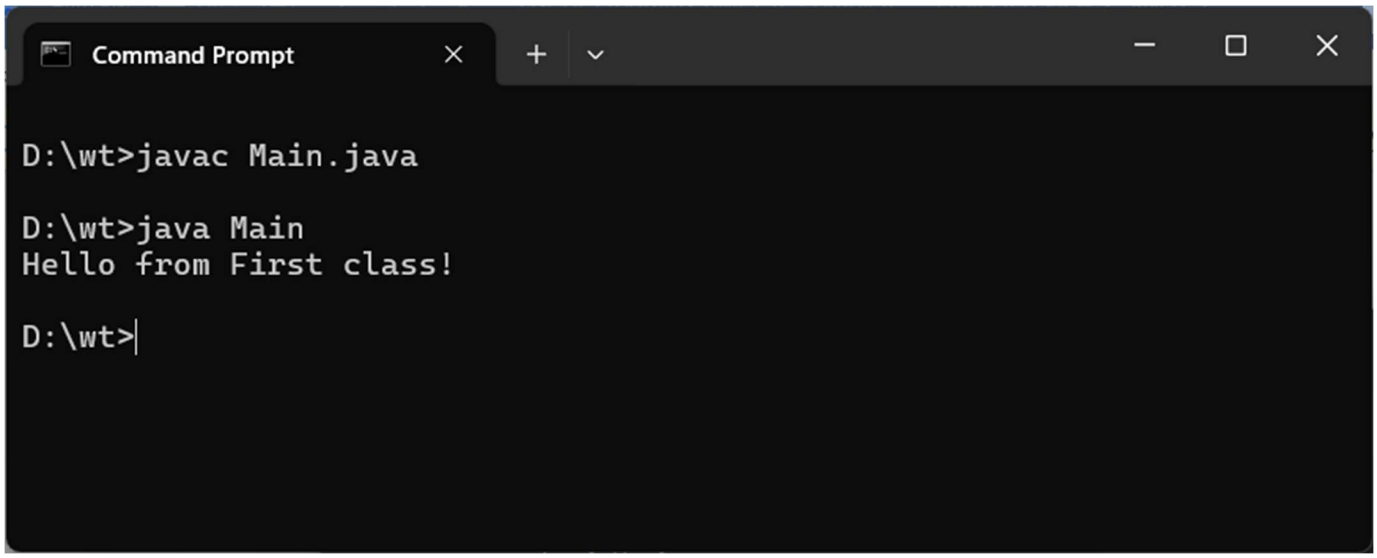
Program No. 3

Objective: Write a program in Java with two classes create a object of first class and call into another class (having main method).

Code Implementation (Source Code):

```
class First {
    void display() {
        System.out.println("Hello from First class!");
    }
}


public class Main {
    public static void main(String[] args) {
        First obj = new First();
        obj.display();
    }
}
```

Output Screen:



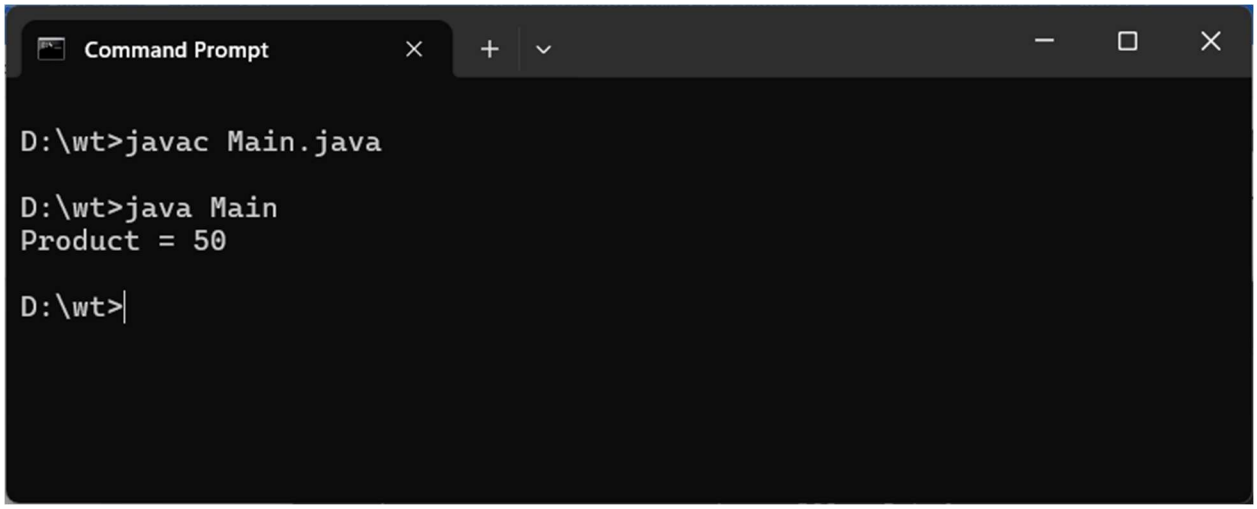
Program No. 4

Objective: Write a program in java to product of two numbers.

Code Implementation (Source Code):

```
public class Main {
    public static void main(String[] args) {
        int a = 5, b = 10;
        int product = a * b;
        System.out.println("Product = " + product);
    }
}
```

Output Screen:



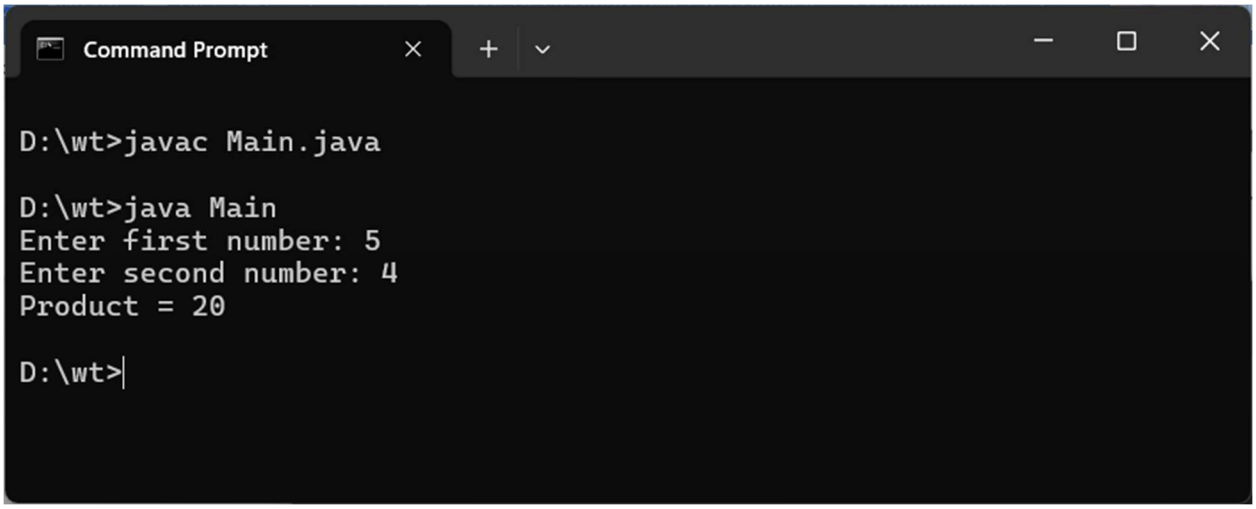
Program No. 5

Objective: Write a program in java product of two numbers (Input by the user).

Code Implementation (Source Code):

```
import java.util.Scanner;


public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first number: ");
        int a = sc.nextInt();
        System.out.print("Enter second number: ");
        int b = sc.nextInt();
        System.out.println("Product = " + (a * b));
    }
}
```

Output Screen:



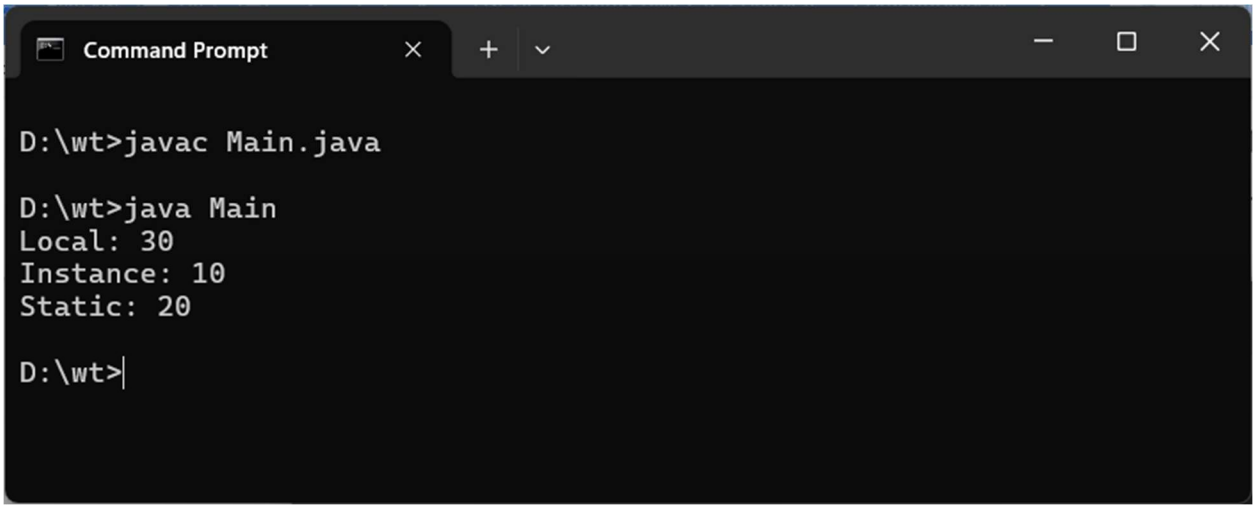
Program No. 6

Objective: Write a program in java to illustrate the concept of local, instance and static variable.

Code Implementation (Source Code):

```
public class Main {
    int instanceVar = 10;
    static int staticVar = 20;

    void show() {
        int localVar = 30;
        System.out.println("Local: " + localVar);
        System.out.println("Instance: " + instanceVar);
        System.out.println("Static: " + staticVar);
    }

    public static void main(String[] args) {
        Main obj = new Main();
        obj.show();
    }
}
```

Output Screen:


Command Prompt
D:\wt>javac Main.java
D:\wt>java Main
Local: 30
Instance: 10
Static: 20
D:\wt>

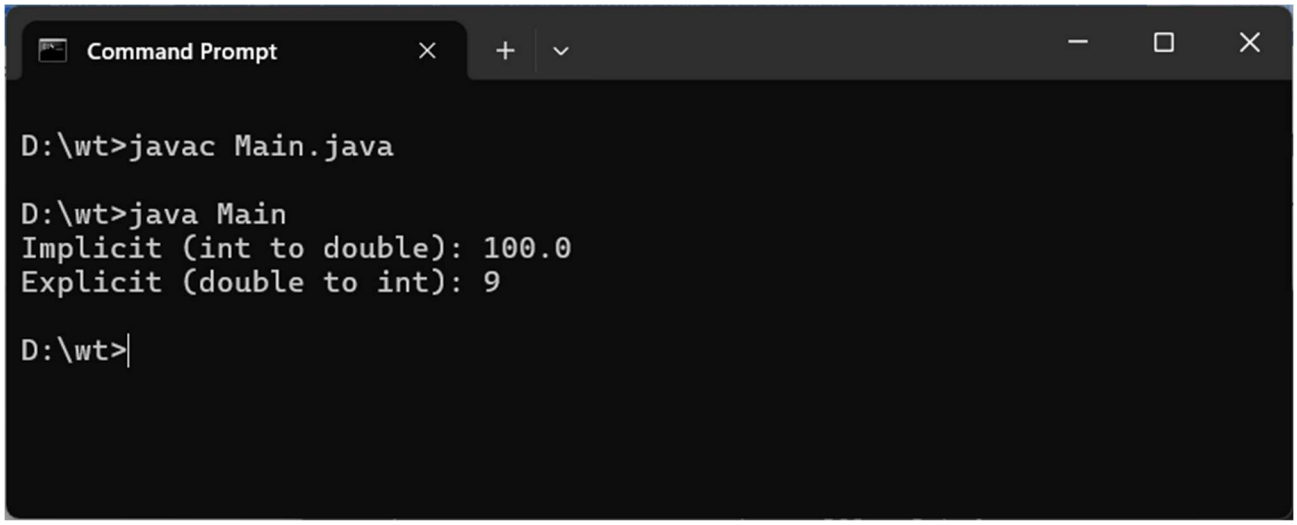
Program No. 7

Objective: Write a program in java to implement implicit and explicit type casting.

Code Implementation (Source Code):

```
public class Main {
    public static void main(String[] args) {
        int i = 100;
        double d = i;
        System.out.println("Implicit (int to double): " + d);

        double x = 9.99;
        int y = (int) x;
        System.out.println("Explicit (double to int): " + y);
    }
}
```

Output Screen:



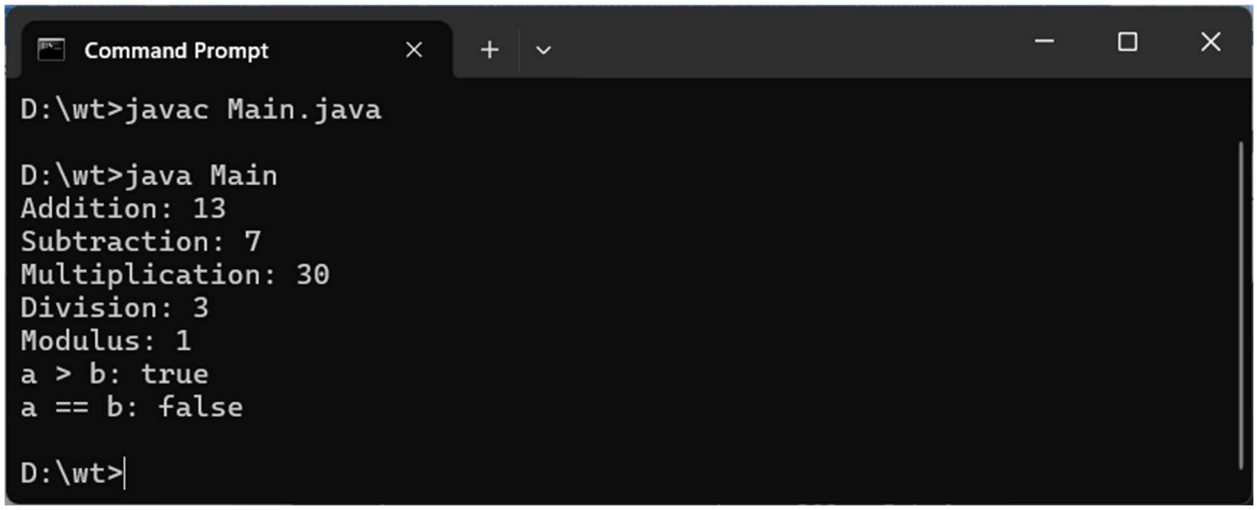
Program No. 8

Objective: Write a program in java to implement the various operators in java.

Code Implementation (Source Code):

```
public class Main {
    public static void main(String[] args) {
        int a = 10, b = 3;
        System.out.println("Addition: " + (a + b));
        System.out.println("Subtraction: " + (a - b));
        System.out.println("Multiplication: " + (a * b));
        System.out.println("Division: " + (a / b));
        System.out.println("Modulus: " + (a % b));
        System.out.println("a > b: " + (a > b));
        System.out.println("a == b: " + (a == b));
    }
}
```

Output Screen:



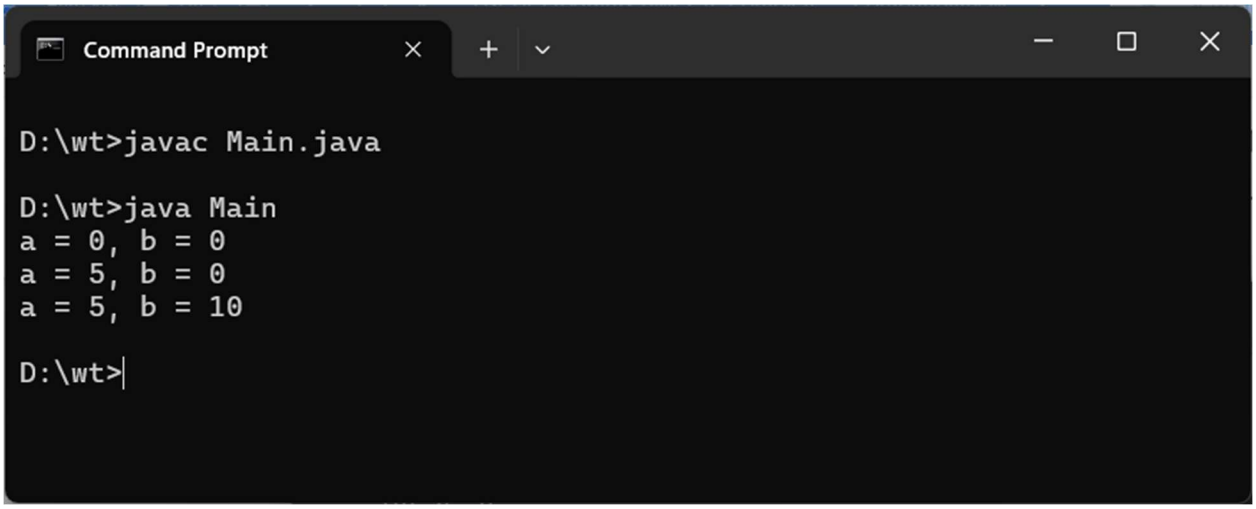
Program No. 9

Objective: Write a program in Java for constructor overloading.

Code Implementation (Source Code):

```
public class Main {
   int a, b;
   Main() {
      a = 0; b = 0;
   }
   Main(int x) {
      a = x;
   }
   Main(int x, int y) {
      a = x; b = y;
   }

   void display() {
      System.out.println("a = " + a + ", b = " + b);
   }

   public static void main(String[] args) {
      new Main().display();
      new Main(5).display();
      new Main(5, 10).display();
   }
}
```

Output Screen:


Command Prompt
D:\wt>javac Main.java
D:\wt>java Main
a = 0, b = 0
a = 5, b = 0
a = 5, b = 10
D:\wt>

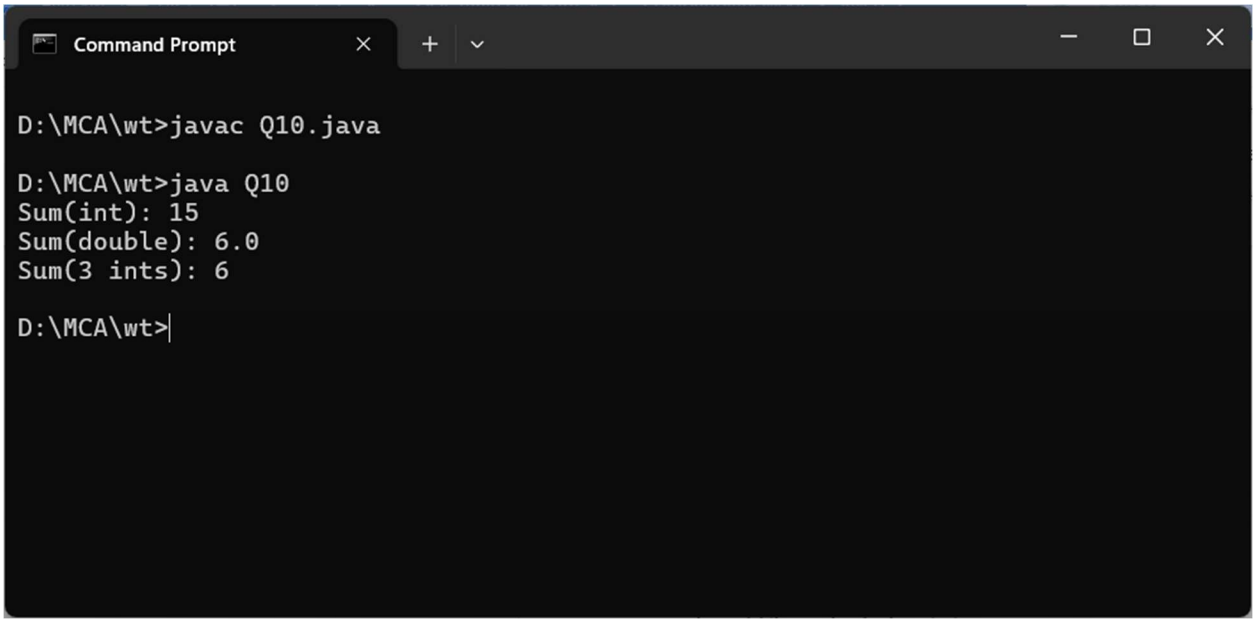
Program No. 10

Objective: Write a program in Java for method overloading.

Code Implementation (Source Code):

```
public class Q10 {
    int add(int a, int b) {
        return a + b;
    }

    double add(double a, double b) {
        return a + b;
    }

    int add(int a, int b, int c) {
        return a + b + c;
    }

    public static void main(String[] args) {
        Q10 obj = new Q10();
        System.out.println("Sum(int): " + obj.add(5, 10));
        System.out.println("Sum(double): " + obj.add(2.5, 3.5));
        System.out.println("Sum(3 ints): " + obj.add(1, 2, 3));
    }
}
```

Output Screen:


Command Prompt
D:\MCA\wt>javac Q10.java
D:\MCA\wt>java Q10
Sum(int): 15
Sum(double): 6.0
Sum(3 ints): 6
D:\MCA\wt>

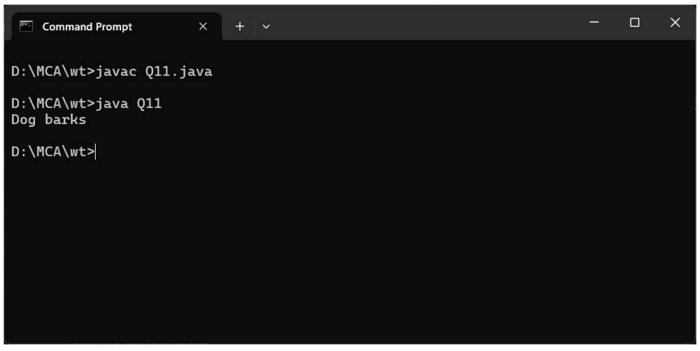
Program No. 11

Objective: Write a program in Java for method overriding.

Code Implementation (Source Code):

```
class Animal {
    void sound() {
        System.out.println("Animal makes a sound");
    }
}
class Dog extends Animal {
    @Override
    void sound() {
        System.out.println("Dog barks");
    }
}
public class Q11 {
    public static void main(String[] args) {
        Animal a = new Dog();
        a.sound();
    }
}
```

Output Screen:



Program No. 12

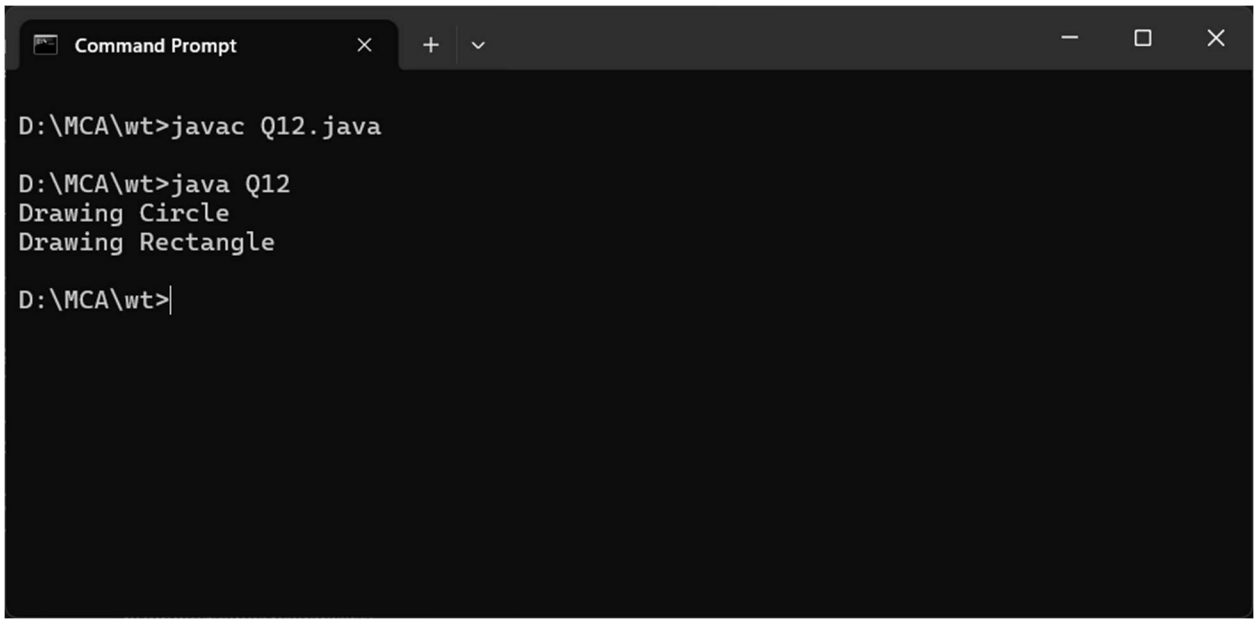
Objective: Write a program in java to show run time polymorphism (up casting).

Code Implementation (Source Code):

```
class Shape {
   void draw() {
      System.out.println("Drawing Shape");
   }
}

class Circle extends Shape {
   void draw() {
      System.out.println("Drawing Circle");
   }
}

class Rectangle extends Shape {
   void draw() {
      System.out.println("Drawing Rectangle");
   }
}

public class Q12 {
   public static void main(String[] args) {
      Shape s;
      s = new Circle();
      s.draw();
      s = new Rectangle();
```

```
        s.draw();
    }
}
```

Output Screen:



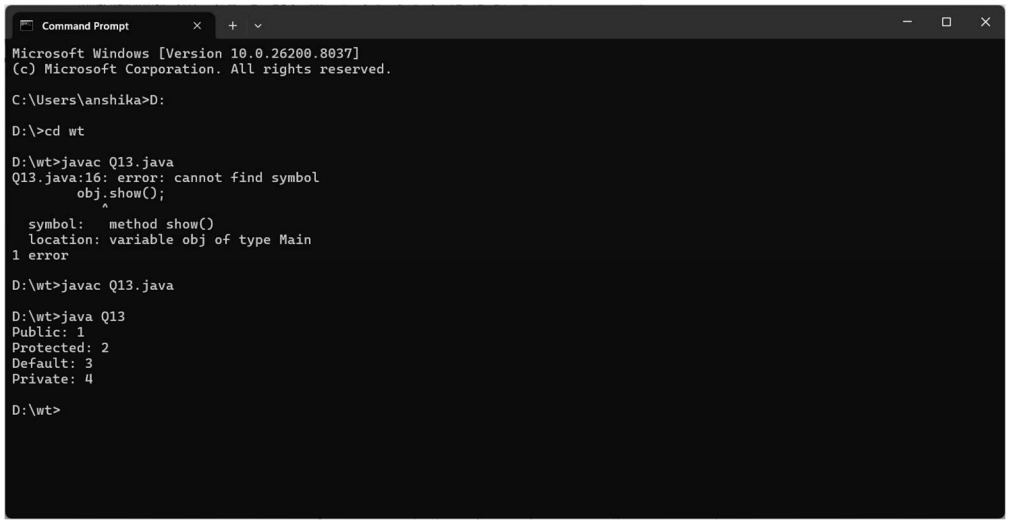
Program No. 13

Objective: Write a program in java for access specifiers (all four).

Code Implementation (Source Code):

```
public class Q13 {
    public int pubVar = 1;
    protected int protVar = 2;
    int defaultVar = 3;
    private int privVar = 4;

    void show() {
        System.out.println("Public: " + pubVar);
        System.out.println("Protected: " + protVar);
        System.out.println("Default: " + defaultVar);
        System.out.println("Private: " + privVar);
    }

    public static void main(String[] args) {
        Q13 obj = new Q13();
        obj.show();
    }
}
```

Output Screen:


Command Prompt
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.
C:\Users\anshika>D:
D:\>cd wt
D:\wt>javac Q13.java
Q13.java:16: error: cannot find symbol
obj.show();
^
symbol: method show()
location: variable obj of type Main
1 error
D:\wt>javac Q13.java
D:\wt>java Q13
Public: 1
Protected: 2
Default: 3
Private: 4
D:\wt>

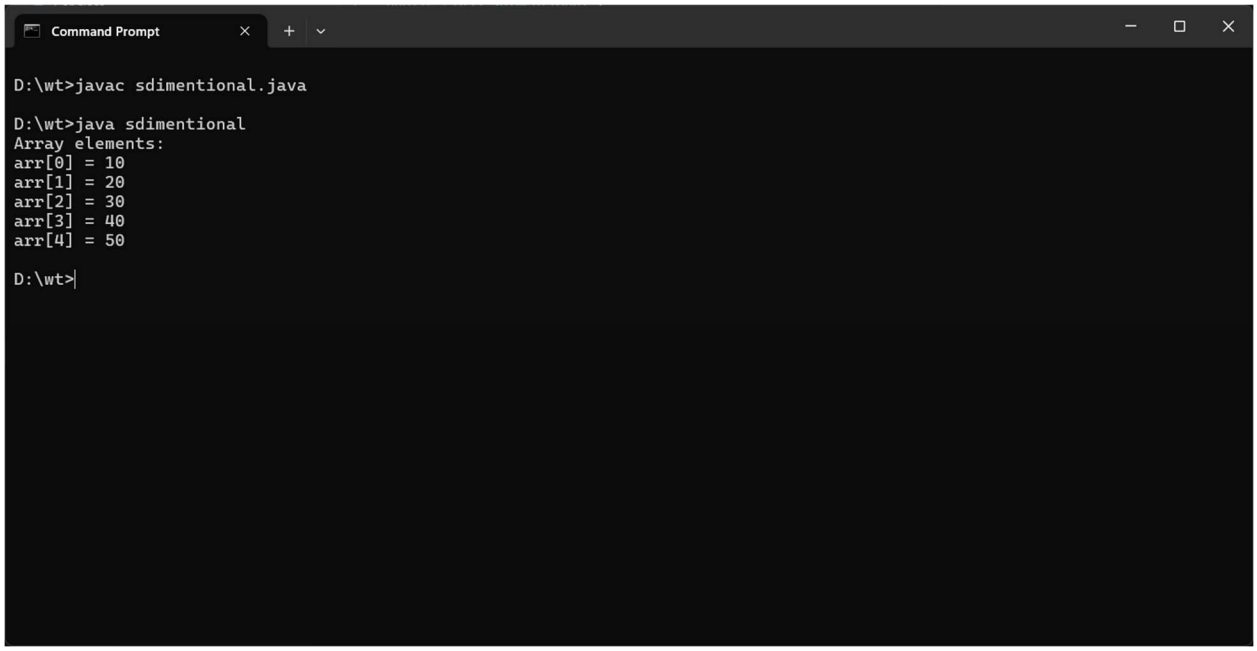
Program No. 14

Objective: Write a program in java to implement the single dimension array.

Code Implementation (Source Code):

```
public class sdimention {
   public static void main(String[] args) {
      int[] arr = {10, 20, 30, 40, 50};
      System.out.println("Array elements:");
      for (int i = 0; i < arr.length; i++) {
         System.out.println("arr[" + i + "] = " + arr[i]);
      }
   }
}
```

Output Screen:



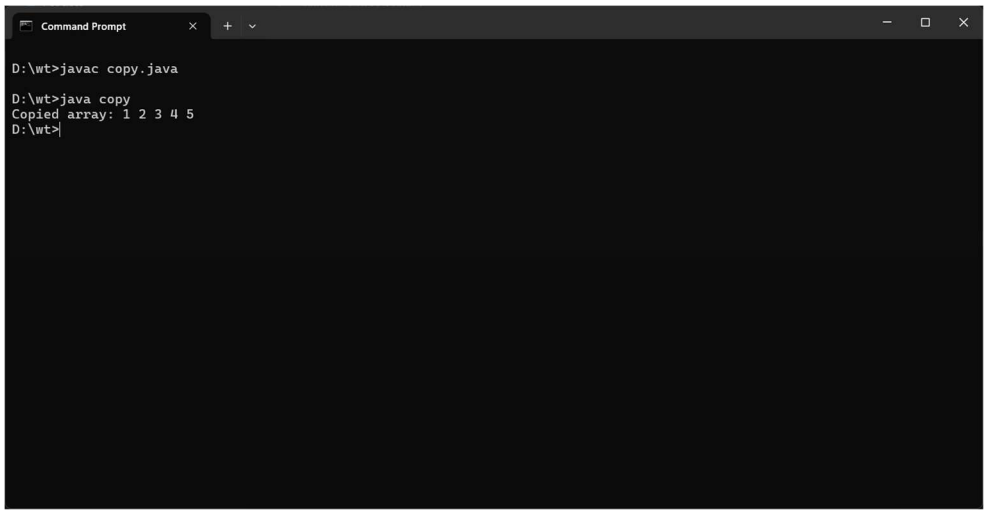
Program No. 15

Objective: Write a program in java to copy the elements from one array to another array.

Code Implementation (Source Code):

```
public class copy {
    public static void main(String[] args) {
        int[] src = {1, 2, 3, 4, 5};
        int[] dest = new int[src.length];
        for (int i = 0; i < src.length; i++) {
            dest[i] = src[i];
        }
        System.out.print("Copied array: ");
        for (int x : dest) {
            System.out.print(x + " ");
        }
    }
}
```

Output Screen:



Program No. 16

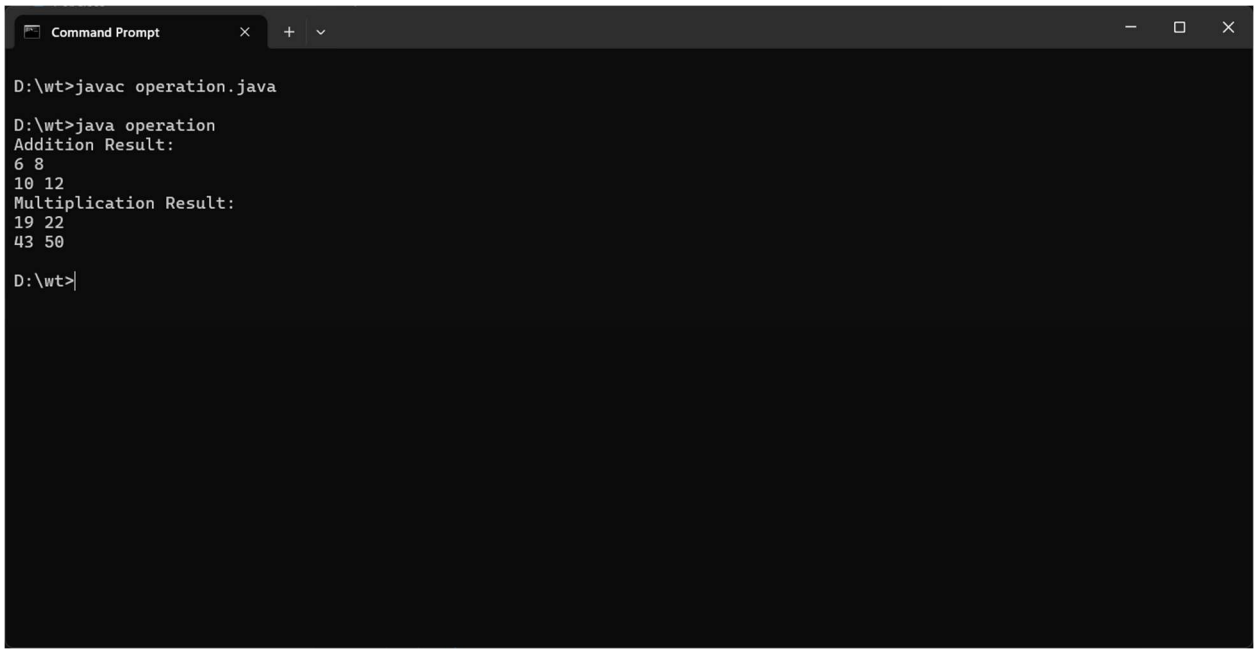
Objective: Write a program in java to perform the addition and multiplication in 2-D array.

Code Implementation (Source Code):

```
public class operation {
   public static void main(String[] args) {
      int[][] a = {{1, 2}, {3, 4}};
      int[][] b = {{5, 6}, {7, 8}};
      int[][] sum = new int[2][2];
      int[][] mul = new int[2][2];

      for (int i = 0; i < 2; i++) {
         for (int j = 0; j < 2; j++) {
            sum[i][j] = a[i][j] + b[i][j];
         }
      }

      for (int i = 0; i < 2; i++) {
         for (int j = 0; j < 2; j++) {
            for (int k = 0; k < 2; k++) {
               mul[i][j] += a[i][k] * b[k][j];
            }
         }
      }

      System.out.println("Addition Result:");
      for (int[] row : sum) {
```

```
            for (int val : row) System.out.print(val + " ");
            System.out.println();
        }
        System.out.println("Multiplication Result:");
        for (int[] row : mul) {
            for (int val : row) System.out.print(val + " ");
            System.out.println();
        }
    }
}
```

Output Screen:



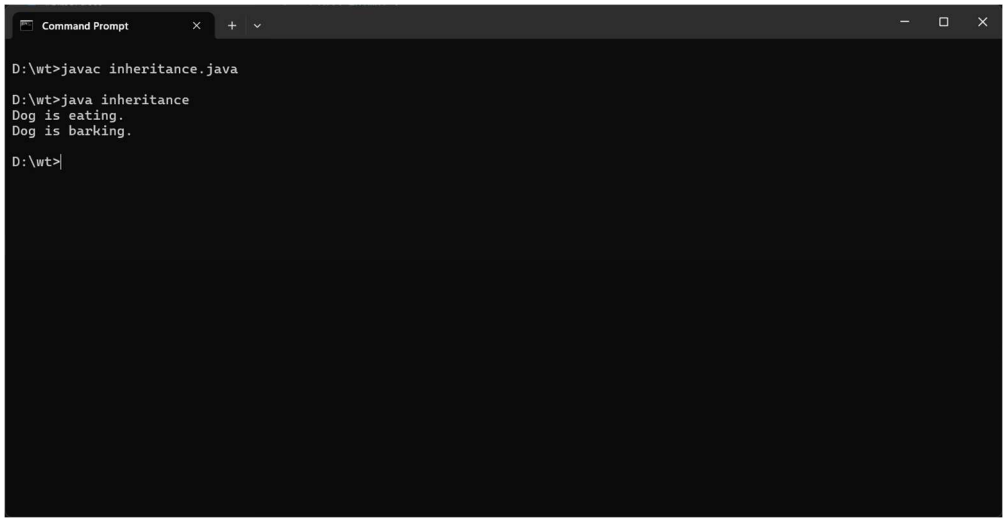
Program No. 17

Objective: Write a program in Java for simple inheritance.

Code Implementation (Source Code):

```
class Animal {
   String name = "Animal";
   void eat() {
      System.out.println(name + " is eating.");
   }
}

class Dog extends Animal {
   void bark() {
      System.out.println("Dog is barking.");
   }
}

public class inheritance {
   public static void main(String[] args) {
      Dog d = new Dog();
      d.name = "Dog";
      d.eat();
      d.bark();
   }
}
```

Output Screen:


Command Prompt
D:\wt>javac inheritance.java
D:\wt>java inheritance
Dog is eating.
Dog is barking.
D:\wt>

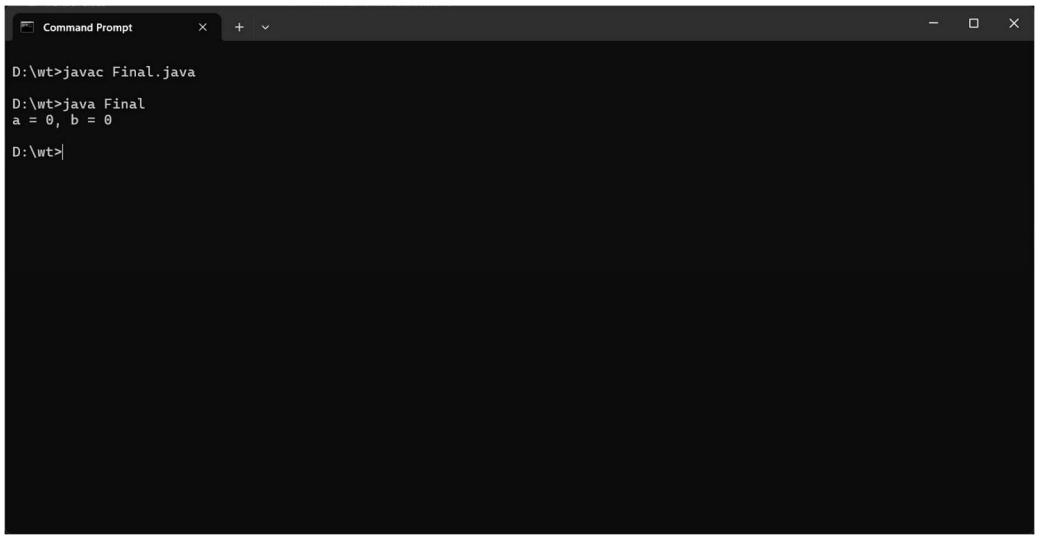
Program No. 18

Objective: Write a program in java for Final keyword.

Code Implementation (Source Code):

```
public class Final {
    final int MAX = 100;

    final void display() {
        System.out.println("Final method called.");
        System.out.println("Final variable MAX = " + MAX);
    }

    public static void main(String[] args) {
        Main obj = new Main();
        obj.display();
    }
}
```

Output Screen:



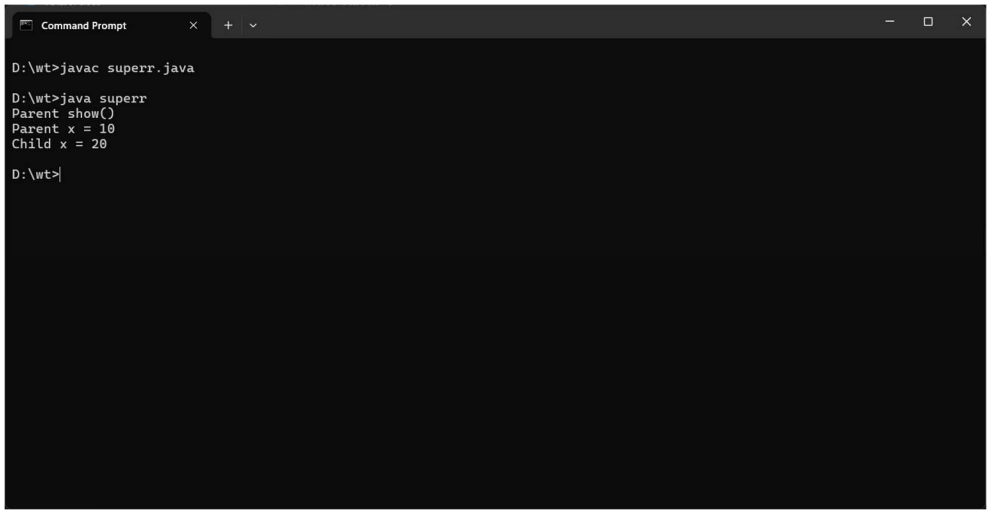
Program No. 19

Objective: Write a program in java for super keyword.

Code Implementation (Source Code):

```
class Parent {
    int x = 10;
    void show() {
        System.out.println("Parent show()");
    }
}

class Child extends Parent {
    int x = 20;
    void show() {
        super.show();
        System.out.println("Parent x = " + super.x);
        System.out.println("Child x = " + x);
    }
}

public class superr {
    public static void main(String[] args) {
        Child c = new Child();
        c.show();
    }
}
```

Output Screen:


Command Prompt
D:\wt>javac superr.java
D:\wt>java superr
Parent show()
Parent x = 10
Child x = 20
D:\wt>

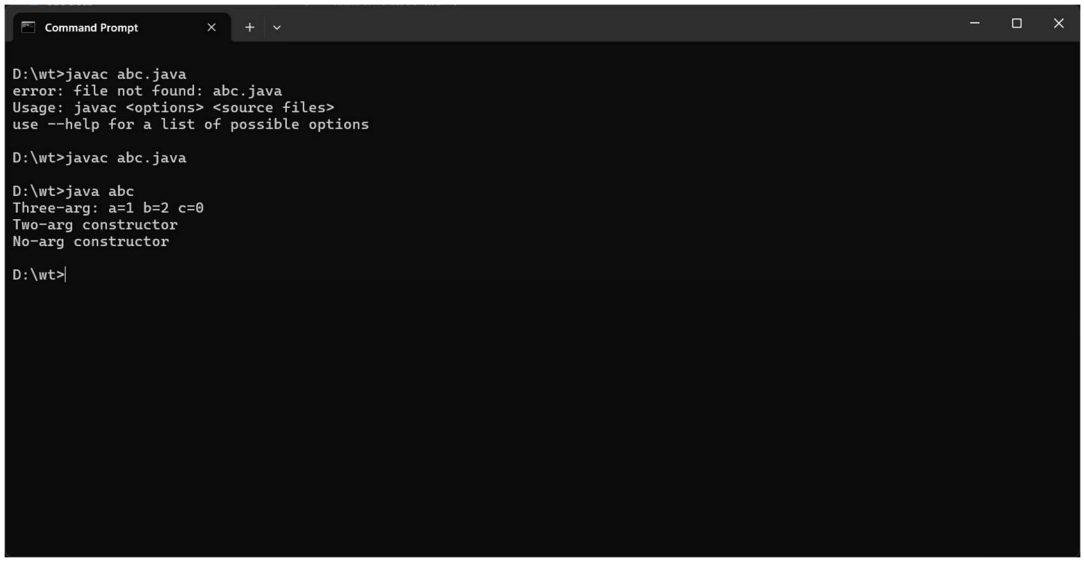
Program No. 20

Objective: Write a program in Java chaining constructor.

Code Implementation (Source Code):

```
public class abc {
   int a, b, c;
   abc() {
      this(1, 2);
      System.out.println("No-arg constructor");
   }
   abc(int a, int b) {
      this(a, b, 0);
      System.out.println("Two-arg constructor");
   }
   abc(int a, int b, int c) {
      this.a = a; this.b = b; this.c = c;
      System.out.println("Three-arg: a=" + a + " b=" + b + " c=" + c);
   }
   public static void main(String[] args) {
      new abc();
   }
}
```

Output Screen:


Command Prompt
D:\wt>javac abc.java
error: file not found: abc.java
Usage: javac <options> <source files>
use --help for a list of possible options
D:\wt>javac abc.java
D:\wt>java abc
Three-arg: a=1 b=2 c=0
Two-arg constructor
No-arg constructor
D:\wt>

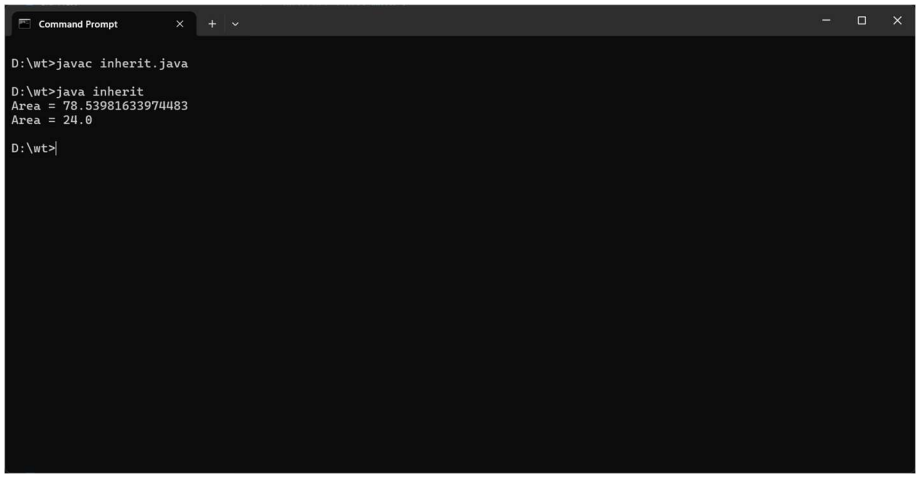
Program No. 21

Objective: Write a program in Java for abstract method and abstract class.

Code Implementation (Source Code):

```
abstract class Shape {
   abstract double area();
   void display() {
      System.out.println("Area = " + area());
   }
}
class Circle extends Shape {
   double r;
   Circle(double r) { this.r = r; }
   double area() { return Math.PI * r * r; }
}
class Rectangle extends Shape {
   double l, w;
   Rectangle(double l, double w) { this.l = l; this.w = w; }
   double area() { return l * w; }
}
public class abstract {
   public static void main(String[] args) {
      Shape s1 = new Circle(5);
      Shape s2 = new Rectangle(4, 6);
      s1.display();
      s2.display();
   }
}
```

Output Screen:



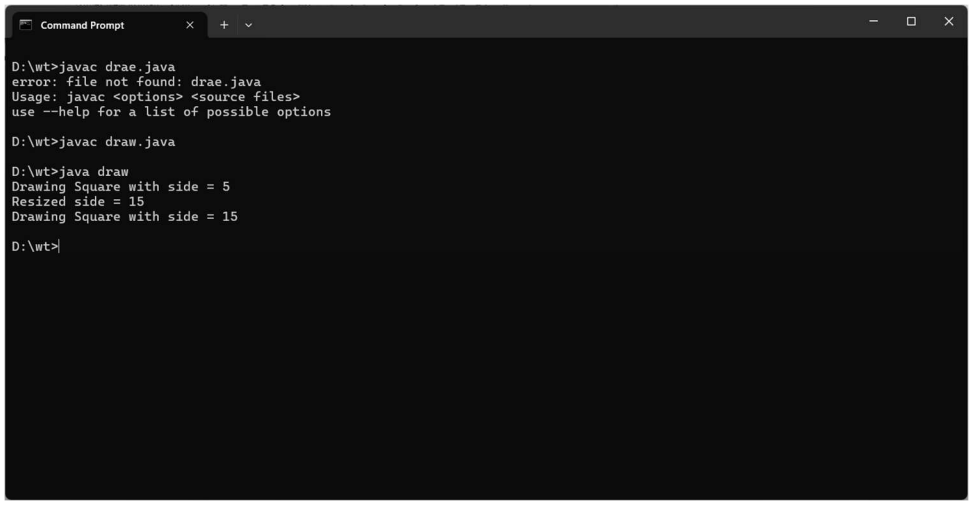
Program No. 22

Objective: Write a program in Java for interface.

Code Implementation (Source Code):

```
interface Drawable {
   void draw();
}

interface Resizable {
   void resize(int factor);
}

class Square implements Drawable, Resizable {
   int side = 5;
   public void draw() {
      System.out.println("Drawing Square with side = " + side);
   }
   public void resize(int factor) {
      side *= factor;
      System.out.println("Resized side = " + side);
   }
}

public class draw {
   public static void main(String[] args) {
      Square sq = new Square();
      sq.draw();
      sq.resize(3);
      sq.draw();
   }
}
```

Output Screen:


Command Prompt
D:\wt>javac drae.java
error: file not found: drae.java
Usage: javac <options> <source files>
use --help for a list of possible options
D:\wt>javac draw.java
D:\wt>java draw
Drawing Square with side = 5
Resized side = 15
Drawing Square with side = 15
D:\wt>

Program No. 23

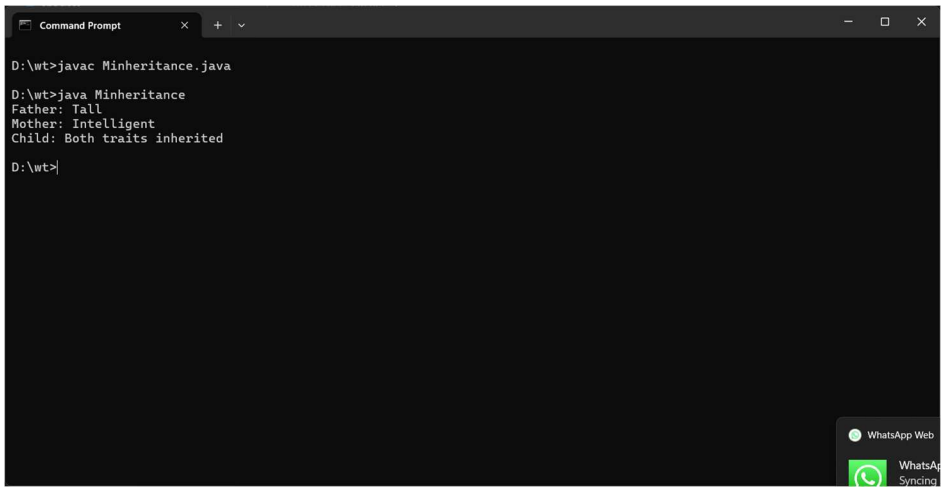
Objective: Write a program in Java multiple inheritance.

Code Implementation (Source Code):

```
interface Father {
   default void tall() {
      System.out.println("Father: Tall");
   }
}

interface Mother {
   default void intelligent() {
      System.out.println("Mother: Intelligent");
   }
}

class Child implements Father, Mother {
   void myTraits() {
      tall();
      intelligent();
      System.out.println("Child: Both traits inherited");
   }
}

public class Minheritance {
   public static void main(String[] args) {
      Child c = new Child();
      c.myTraits();
```

```
    }
}
```

Output Screen:



Program No. 24

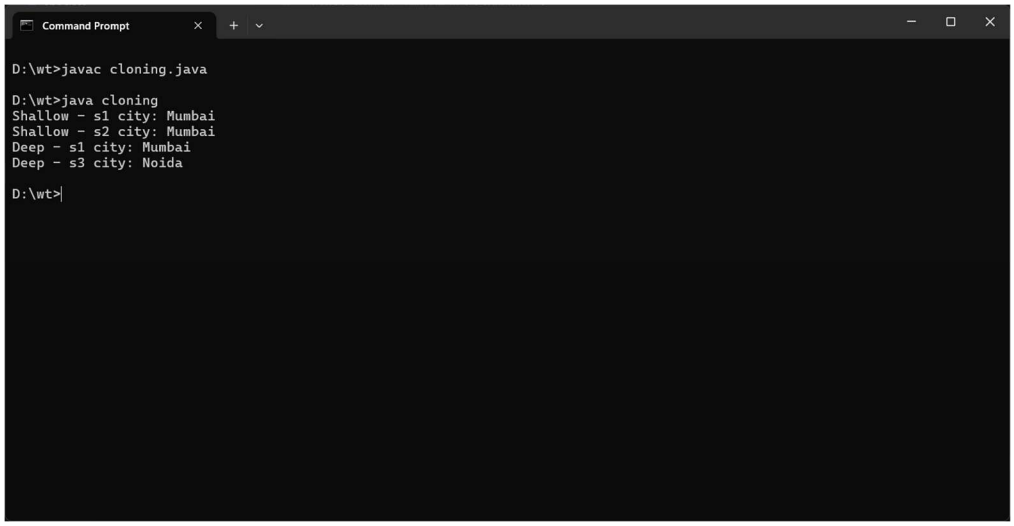
Objective: Write a program in Java for Object Cloning (shallow and deep copy).

Code Implementation (Source Code):

```
class Address implements Cloneable {
    String city;
    Address(String city) { this.city = city; }
    protected Object clone() throws CloneNotSupportedException {
        return super.clone();
    }
}
class Student implements Cloneable {
    String name;
    Address addr;
    Student(String name, Address addr) {
        this.name = name; this.addr = addr;
    }

    protected Object clone() throws CloneNotSupportedException {
        return super.clone();
    }

    Student deepClone() throws CloneNotSupportedException {
        Student s = (Student) super.clone();
        s.addr = (Address) addr.clone();
        return s;
    }
}
```

```
public class cloning {
    public static void main(String[] args) throws CloneNotSupportedException {
        Address a = new Address("Delhi");
        Student s1 = new Student("Manu", a);
        Student s2 = (Student) s1.clone();
        Student s3 = s1.deepClone();

        s2.addr.city = "Mumbai";
        System.out.println("Shallow - s1 city: " + s1.addr.city);
        System.out.println("Shallow - s2 city: " + s2.addr.city);

        s3.addr.city = "Noida";
        System.out.println("Deep - s1 city: " + s1.addr.city);
        System.out.println("Deep - s3 city: " + s3.addr.city);
    }
}
```

Output Screen:



Program No. 25

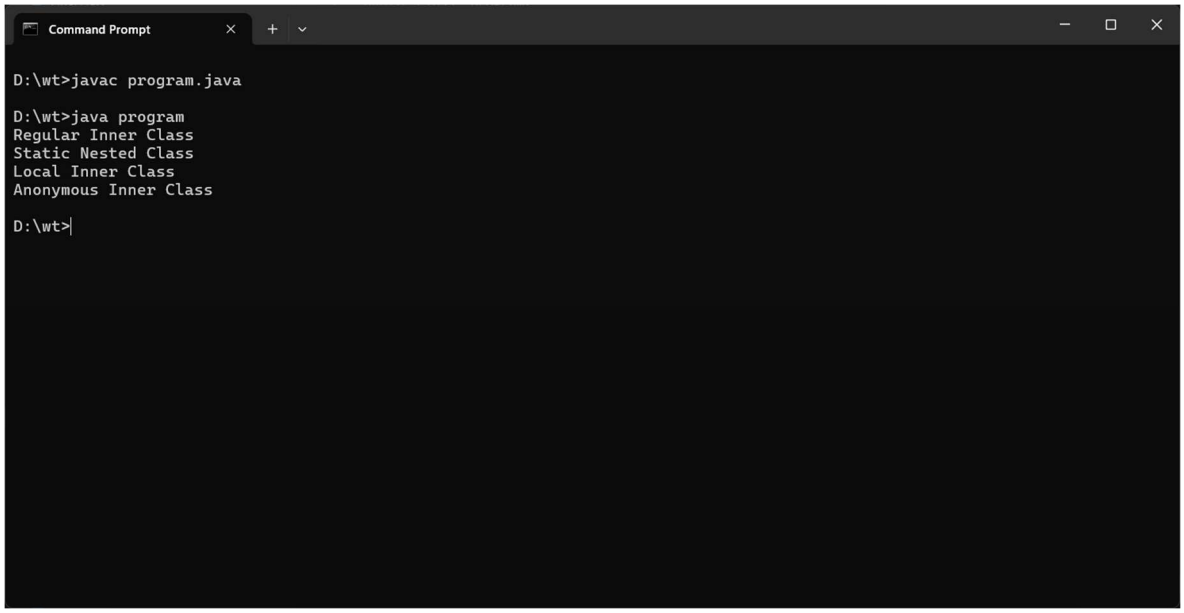
Objective: Write a program in Java for Inner Classes (all types).

Code Implementation (Source Code):

```
public class  program{
   class Inner {
      void show() { System.out.println("Regular Inner Class"); }
   }
   static class StaticNested {
      void show() { System.out.println("Static Nested Class"); }
   }

   void methodWithLocalClass() {
      class Local {
         void show() { System.out.println("Local Inner Class"); }
      }
      new Local().show();
   }

   public static void main(String[] args) {
      program outer = new program();
      outer.new Inner().show();
      new StaticNested().show();


      outer.methodWithLocalClass();
```

```
        Runnable r = new Runnable() {
            public void run() { System.out.println("Anonymous Inner Class"); }
        };
        r.run();
    }
}
```

Output Screen:



Program No. 26

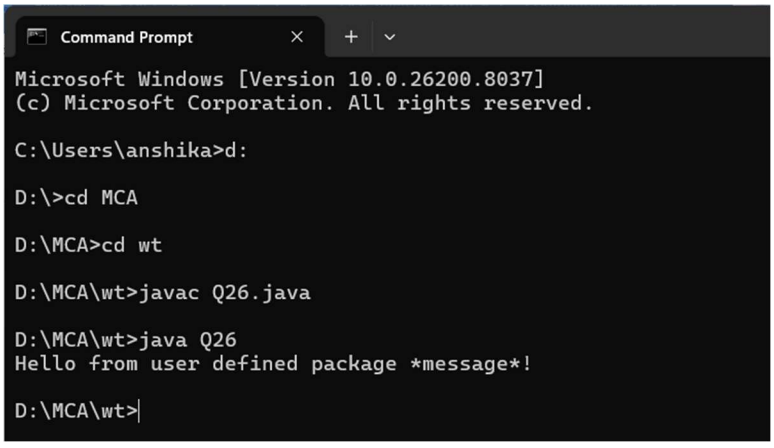
Objective: Write a program in java to create the package (user defined package).

Code Implementation (Source Code):

```
package myPackage;

public class Message {
    public void show() {
        System.out.println("Hello from user defined package *message*!");
    }
}
```

```
import myPackage.Message;

public class Q26 {
    public static void main(String[] args) {
        Message msg = new Message();
        msg.show();
    }
}
```

Output Screen:



Program No. 27

Objective: Write a program in Java for exception handling by using try, catch and finally.

Code Implementation (Source Code):

```
public class Q27 {
    public static void main(String[] args) {
        try {
            int a = 10, b = 0;
            int result = a / b;
            System.out.println("Result = " + result);
        } catch (ArithmeticException e) {
            System.out.println("Exception caught: " + e.getMessage());
        } finally {
            System.out.println("Finally block always executes.");
        }
    }
}
```

Output Screen: